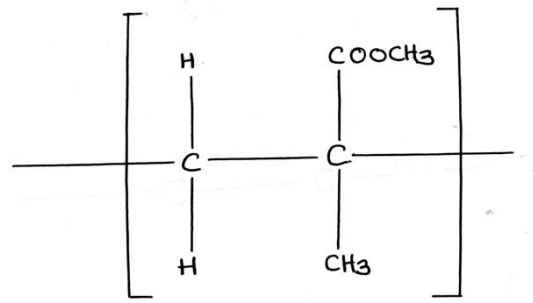


Question				Marking details	Marks available					
					AO1	AO2	AO3	Maths	Prac	Total
12.	(a)	(i)		electrophilic addition		1				1
		(ii)		acidified (potassium) dichromate / H ⁺ , Cr ₂ O ₇ ²⁻ / acidified potassium manganate(VII) / H ⁺ , MnO ₄ ⁻	1				1	1
		(iii)		ZnBr ₂			1			1
	(b)	(i)		⁻ CN		1				1
		(ii)		HCN is added across the C=O bond		1				1
		(iii)		dilute sulfuric acid / H ₂ SO ₄ (aq) / dilute hydrochloric acid/ HCl(aq) neutral answer – water	1					1
		(iv)		dehydrating agent / removes water	1				1	1
		(v)				1				1

Question				Marking details	Marks available																	
					AO1	AO2	AO3	Maths	Prac	Total												
		(vi)		in condensation polymerisation a small molecule is eliminated which does not occur with addition polymerisation answer must refer clearly to both types of polymerisation	1					1												
	(c)	(i)		2,3,3-trichloropentane		1				1												
		(ii)		<table><tr><td>Hydrogen proton</td><td>Splitting pattern</td></tr><tr><td>a</td><td>triplet</td></tr><tr><td>b</td><td>quartet</td></tr><tr><td>c</td><td>quartet</td></tr><tr><td>d</td><td>doublet</td></tr></table> all four correct (2) two or three correct (1)	Hydrogen proton	Splitting pattern	a	triplet	b	quartet	c	quartet	d	doublet								2
Hydrogen proton	Splitting pattern																					
a	triplet																					
b	quartet																					
c	quartet																					
d	doublet																					
				Total question 12	4	7	1	0	2	12												